# Capstone Project Report
# Predicting High-Performing Firms Using Financial Ratios and Machine Learning

## GitHub Repository

**https://github.com/salscilanga/financial-performance-ml**

## Executive Summary

This project investigated whether financial ratios and company characteristics could be used to predict firm performance using machine learning techniques. Several financial ratios were engineered from the raw dataset, including leverage ratio, cash ratio, PPE ratio, and retained earnings ratio. Exploratory analysis revealed skewed distributions and significant outliers within the financial data.

Logistic Regression, Random Forest, and SVM models were evaluated using accuracy, precision, and recall metrics. Logistic Regression with PCA achieved the strongest balance between predictive performance and interpretability, achieving approximately 93% recall while maintaining acceptable accuracy and precision.

Feature importance analysis showed that firm size was the strongest predictor of high performance. Cash ratio exhibited a stronger negative relationship with performance compared to leverage and PPE ratio, suggesting that firms holding excessive idle liquidity were less likely to be classified as high-performing.

## Introduction

The objective of this project was to determine whether financial variables and company characteristics could be used to identify high-performing firms. The project combined financial analysis, exploratory data analysis, statistical testing, and machine learning in order to develop a predictive classification framework.

A data-driven solution was considered appropriate because machine learning methods are capable of identifying relationships and patterns that may not be visible using traditional statistical techniques alone.

## Methods

The project began with a raw financial dataset containing accounting variables and company information. Data preprocessing included reviewing data types, handling missing values, and separating numerical and categorical variables.

Feature engineering was performed to create interpretable financial indicators such as leverage ratio, cash ratio, PPE ratio, equity ratio, and retained earnings ratio.

Exploratory Data Analysis (EDA) included:

• Correlation analysis
• Distribution analysis
• Histograms and heatmaps
• Statistical hypothesis testing

Because the financial variables displayed skewness and outliers, non-parametric statistical methods were used. Logistic Regression, Random Forest, and Support Vector Machine (SVM) models were evaluated using accuracy, precision, and recall metrics.

### Hyperparameter Tuning

To improve predictive performance, hyperparameter tuning was performed for the machine learning models using GridSearchCV.

Several parameter combinations were evaluated for Logistic Regression, Random Forest, and Support Vector Machine (SVM) models using cross-validation on the training dataset.

The tuning process aimed to identify parameter configurations that maximized predictive performance while reducing overfitting.

Examples of tuned parameters included:

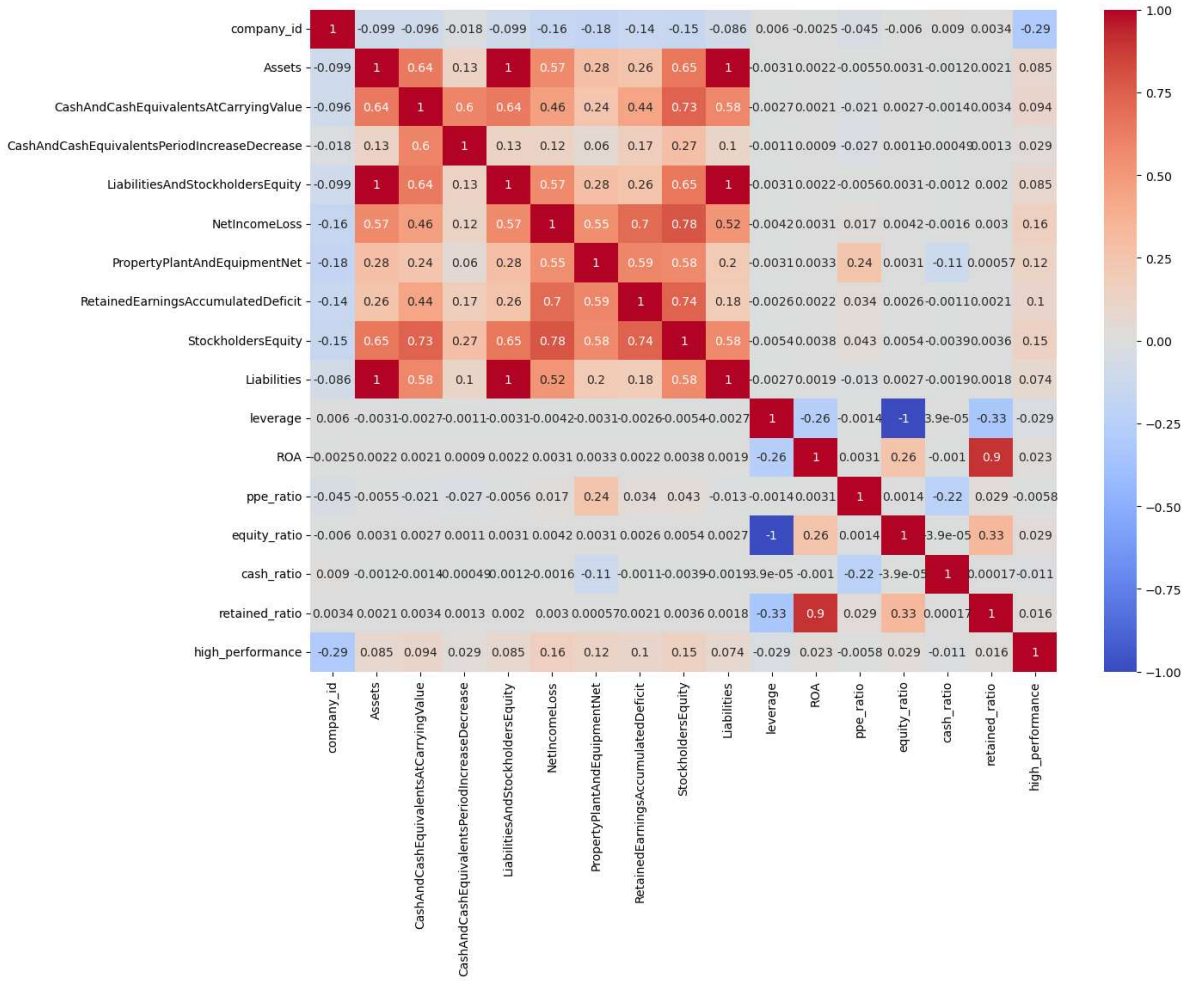
- regularization strength for Logistic Regression,
- number of estimators and tree depth for Random Forest,
- and kernel parameters for SVM.

The optimized models were subsequently evaluated on the validation and test datasets using accuracy, precision, and recall metrics.

## Exploratory Data Analysis

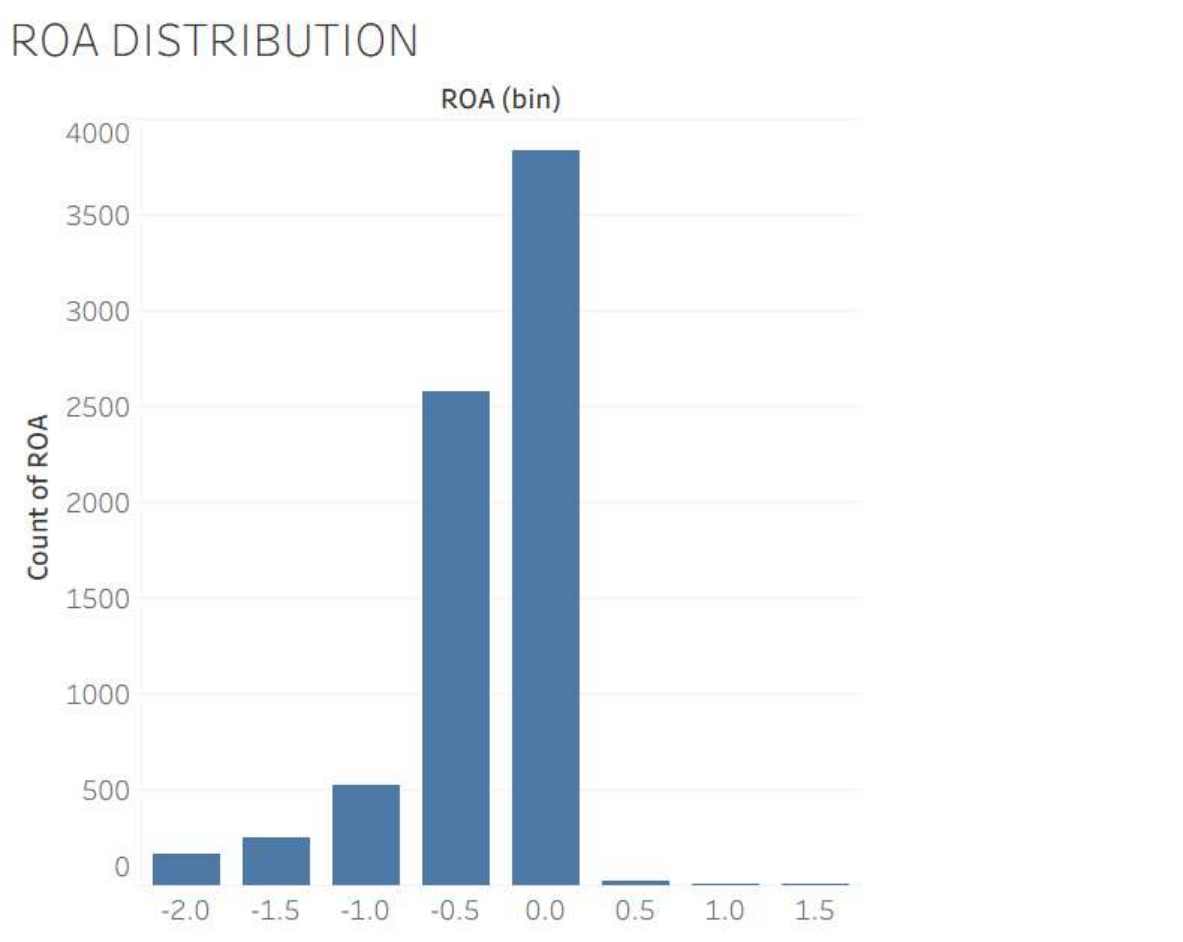
The exploratory analysis identified skewed financial distributions, significant outliers, and relationships between financial ratios and firm performance.

### Figure 1: Correlation Heatmap




The heatmap shows relationships between financial variables and engineered ratios. Cash ratio and leverage exhibited negative relationships with profitability.

### Figure 2: ROA Distribution




The ROA distribution exhibited strong skewness and extreme outliers, with most observations concentrated around small profitability values.

## Statistical Hypothesis Testing

To determine whether financial characteristics differed significantly between high-performing and low-performing firms, non-parametric statistical testing was conducted. Because the financial variables displayed skewness and outliers, Mann–Whitney U tests were used instead of traditional t-tests for continuous variables.

The statistical tests showed significant differences between performance groups for leverage ratio, cash ratio, and PPE ratio. A Chi-square test also identified a strong relationship between company size and firm performance.

| Variable | Test | P-value |
|---|---|---|
| Leverage Ratio | Mann–Whitney U | < 0.001 |

| | | |
|---|---|---|
| Cash Ratio | Mann–Whitney U | < 0.001 |
| PPE Ratio | Mann–Whitney U | 0.0015 |
| Company Size | Chi-square | < 0.001 |

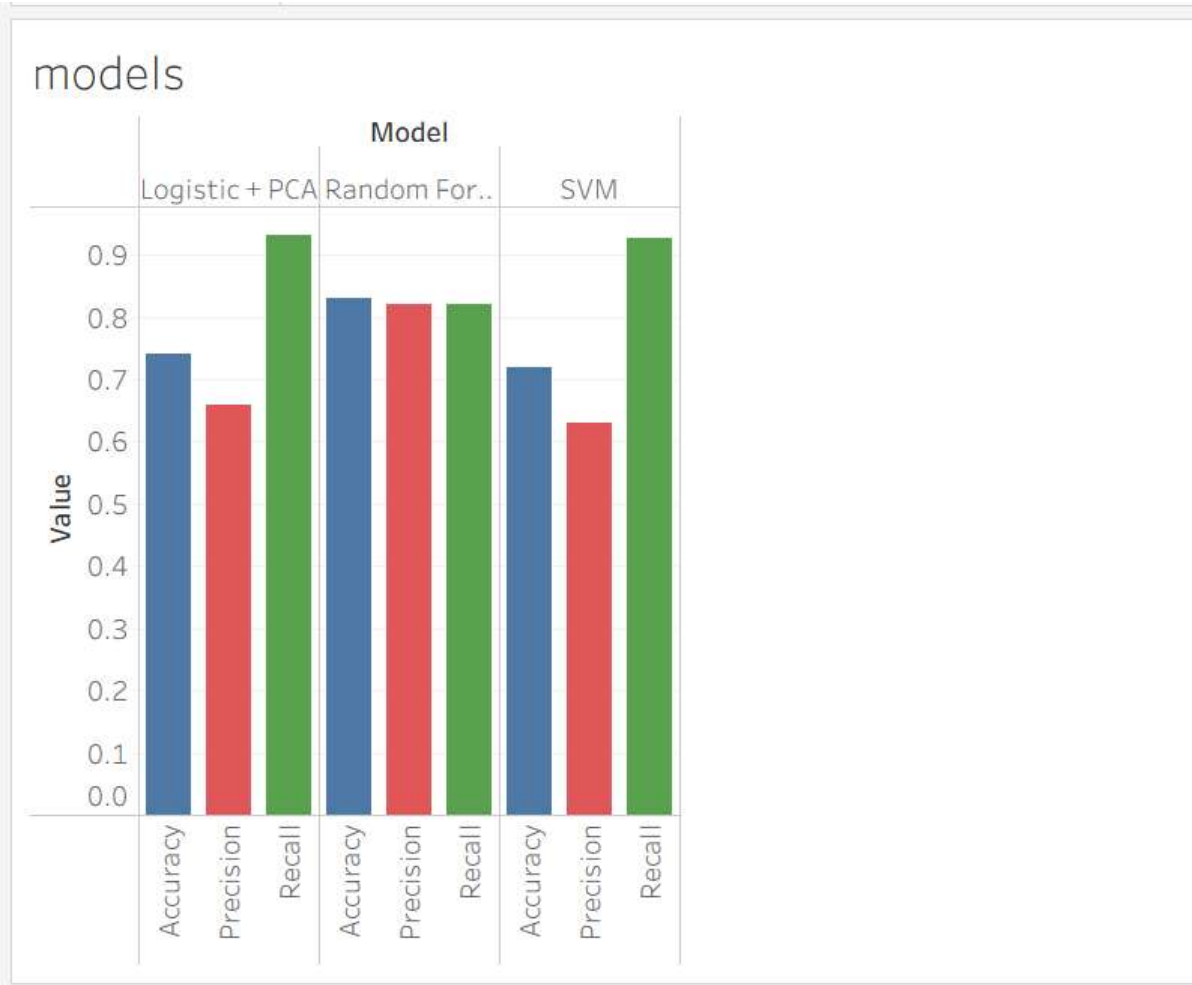
## Results and Model Evaluation

Random Forest achieved the highest overall accuracy and precision, while Logistic Regression with PCA achieved the highest recall. Logistic Regression with PCA was selected as the final model because minimizing false negatives aligned more closely with the business objective.

Feature importance analysis indicated that firm size was the strongest positive predictor of high performance. Cash ratio showed a stronger negative effect than leverage and PPE ratio, suggesting that firms with excessively high liquidity may not be deploying resources efficiently.

| Model | Accuracy | Precision | Recall |
|---|---|---|---|
| Logistic Regression + PCA | 0.74 | 0.66 | 0.931 |
| Random Forest | 0.83 | 0.82 | 0.82 |
| SVM | 0.72 | 0.63 | 0.928 |

## Figure 3: Machine Learning Model Comparison



## Figure 4: Feature Impact on High Performance

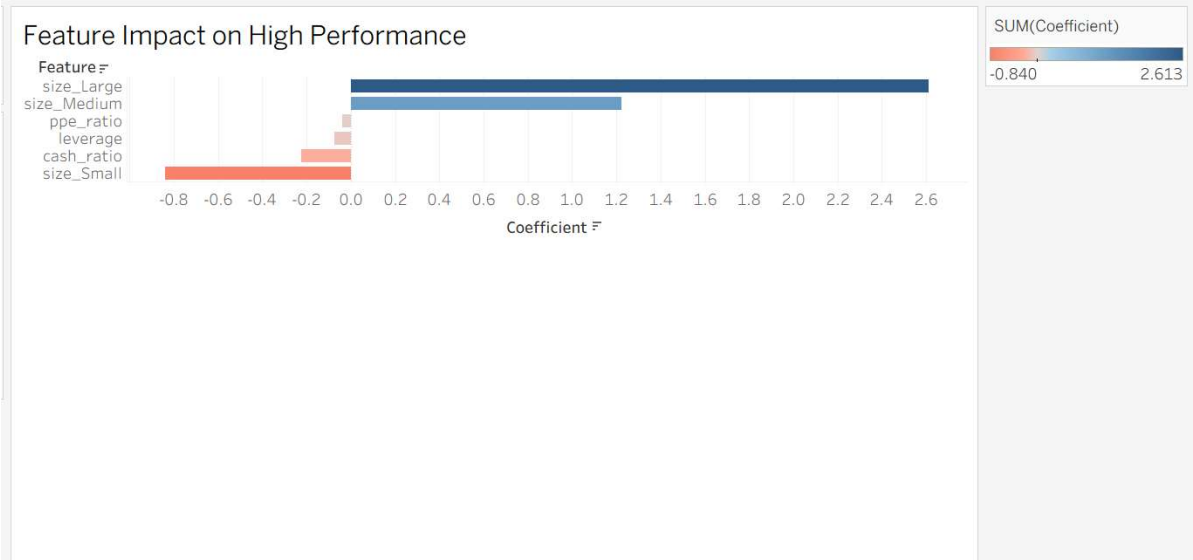
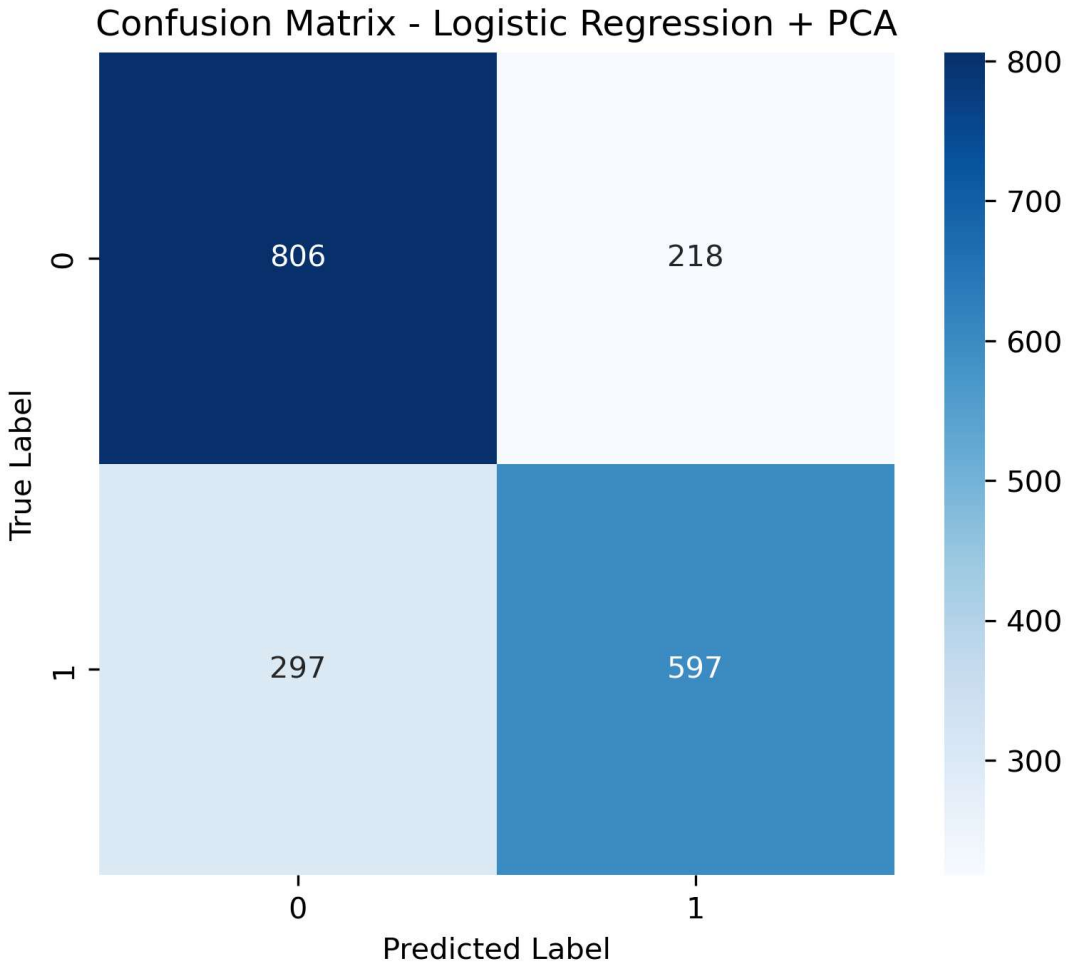


**Figure 5: Confusion Matrix - Logistic Regression + PCA**




## Business Interpretation and Cost of Misclassification

Median ROA analysis showed a substantial profitability gap between high-performing and low-performing firms. The median ROA for high-performing firms was approximately 2.88%, compared to approximately -55.7% for low-performing firms, representing an opportunity gap of roughly 58.6 percentage points.

This suggests that incorrectly failing to identify a high-performing firm may result in significant missed profitability opportunities, while selecting a poorly performing company may expose investors to materially weaker financial outcomes. Consequently, minimizing false negatives was considered strategically more important than maximizing overall accuracy.

## Conclusion

This project demonstrated that machine learning techniques can be effectively applied to financial data in order to classify firm performance. The analysis showed that firm size was the strongest explanatory factor, while cash ratio exhibited a stronger negative relationship with performance compared to leverage and PPE ratio.

Logistic Regression with PCA emerged as the most suitable final model because it achieved the highest recall while preserving interpretability. The project also demonstrated the importance of robust preprocessing and statistical analysis when working with real-world financial datasets containing skewness and outliers.